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UNIT: Career Development Cell (CDC) - Directorate Office	**SECURITY:** STRICTLY CONFIDENTIAL • INTERNAL
SUBJECT: Official Portal Administrator & Operations Manual	**EFFECTIVE DATE:** Batch of 2026 (Annual Release)

SPSU Career Development Cell (CDC) • Directorate Office

Power BI Executive Intelligence Portal - Administrator Guide

1. Executive Summary & Architecture

The **SPSU Career Development Cell (CDC) Power BI Executive Intelligence Portal** is an institutional decision-support and placement management platform engineered specifically for the **Director, Vice Chancellor, and Career Development Cell leadership**. It delivers real-time visibility into graduating cohort placement conversions, summer internship commitments, compensation benchmarks, and corporate recruiter engagements.

Core Architecture & Design Standards: Constructed on a modern React 19 and Vite 6 framework styled with Microsoft Power BI executive design patterns (fluent header ribbon, dynamic filter slicers, KPI performance cards, drill-through student profile dossiers, and sub-second multi-field query engine).

Component	Technology / Specification	Operational Purpose
Frontend Engine	React 19 + Vite 6 + Tailwind CSS	Sub-second client rendering & responsive layout across desktop and mobile devices.
Design System	Power BI Executive Ribbon Edition	Institutional presentation standards, interactive slicers, and real-time live clock.
Dataset Size	392 Students / 9 Disciplines	CSE (192), ECE (18), MN (17), CE (11), ME (10), BDES (2), BCA (38), BBA (57), MBA (47).
Hosting Topology	GitHub Pages Cloud + Standalone HTML	Dual availability: live cloud HTTPS portal plus 718 KB offline standalone file.
Security Protocol	Role-Gated Authentication Gate	Directorate access protection with session tokens and instant logout revocation.

2. Access Control & Administrator Credentials

Access to the executive intelligence console is strictly restricted through the **SPSU Authentication Gate**. The portal validates incoming credentials against authorized Directorate keys and grants an encrypted session token.

AUTHENTICATION GATE CREDENTIALS	DESIGNATED ROLE: Directorate Executive / Super Admin
Username: **`ADMIN`** (Case-insensitive)	**Password:** **`SPSU123`**
Auto-Fill Shortcut: Click *'Auto-fill Credentials'* for 1-click login.	**Session Persistence:** *'Keep me logged in'* preserves authentication.

Sign-Out & Session Revocation:

- **Direct Sign Out:** Click the red **Log Out icon** in the top-right header ribbon to immediately revoke authentication tokens and lock the portal.
- **Role Switcher Modal:** Click the User Profile badge (*'Administrator / Super Admin'*) to open the Auth Gateway modal to preview department auditor views.

3. Portal Modules & Intelligence Directory

The portal is organized into seven specialized intelligence views accessible via the primary ribbon navigation:

Ribbon Tab	Core Intelligence Capabilities	Primary Target Audience
Executive Cockpit	Consolidated KPI scorecard (Placements, Summer Internships, Peak CTC Rs. 10.20 LPA, Average CTC Rs. 5.85 LPA), uncropped campus accreditation banner, and ticking live clock.	Director, Vice Chancellor & Deans
Placement Hub	FTE vs PPO breakdown (19 FTE / 28 PPO), Tier 1 (>=8 LPA), Tier 2 (5-8 LPA), Tier 3 (<5 LPA) package tiers, unplaced rosters, and company offer metrics.	Placement Officers & CDC Team
Summer Internship	Summer internship conversion tracking (75.8% conversion, 374 cumulative offers), stipend distributions, and male-female cohort parity metrics.	Internship Coordinators
Student 360° Matrix	Granular student database with multi-field instant search, numerical column sorting (CTC, CGPA, Backlogs, S.No), department filtering, and modal dossiers.	Academic Advisors & CDC Staff
Recruiter Intelligence	Tier 1 hiring partners (KMK Venture, Country Edu, Xebia, Celebal, Eleation), offer distribution, salary ranges, and historical corporate drive performance.	Corporate Relations Managers
Reports Studio	Official Director's 1-Pager Executive Memo generator, CSV student datasets, Department Matrices, and print-ready PDF memo formatting.	Executive Assistants & Directorate
Template Manager	Annual rollover data ingestion engine, drag-and-drop CSV master upload, and schema field mapping validation for new academic batches.	Database & IT Administrators

4. Interactive Power BI Slicers & Query Operations

A cornerstone of the executive portal is the **Collapsible Power BI Slicers Pane**, accessible via the **'Slicers'** button in the top ribbon.

- **Discipline / Program Slicer:** Filter all views across 9 degree disciplines (CSE, ECE, MN, CE, ME, BDES, BCA, BBA, MBA) or select 'ALL' for institutional totals.
- **Placement Status Slicer:** Segment cohorts into Placed-FTE, Placed-PPO, Remaining Unplaced, or Opt-Out categories.
- **Summer Internship Slicer:** Isolate students who have secured internships vs. those awaiting industry placement.
- **Instant Global Slicer Reset:** Click the **'Reset'** button in the ribbon to restore default parameters across all 392 students.

Student 360° Matrix - Deep Search & Numerical Sorting:

The Student 360° Matrix is equipped with an optimized query engine built to handle large institutional cohorts without interface lag:

1. **Multi-Field Instant Search:** Type any search term. The query engine matches Student Name, Enrollment Number (e.g. 2223010001), SPSU Email, Department, Specialization (Cloud, AI/ML, Cyber Security), and Recruiter Companies simultaneously.
2. **True Numerical Sorting:** Click column headers (**S.No, CTC, CGPA, Backlogs**) to sort in true mathematical sequence (e.g., Rs. 10.20 LPA → Rs. 8.00 LPA → Rs. 6.50 LPA) rather than lexicographical string sorting.
3. **Automated Pagination Reset:** Whenever a search term or department filter changes, the table automatically resets to Page 1 to eliminate blank-page display states.
4. **Drill-Down Profile Dossier:** Click the **'View Profile'** eye icon next to any student to open their comprehensive 360° Profile Dossier containing verified backlogs, internship offers, company CTC, and contact coordinates.

5. Export Engine & 1-Click Executive Memo Generation

The Directorate portal incorporates an automated multi-format export pipeline to facilitate executive reporting:

Export Option	Output Format	Description & Operational Usage
Director's 1-Pager Memo	**PDF / Print Dialog**	Generates an official executive briefing document with embedded university banner, key KPI metrics, program-wise placement table, and formal signature lines for the Director.
Filtered Students Dataset	**CSV Spreadsheet**	Downloads all current filtered student records with full attributes (Enrollment, CGPA, CTC, Company, Backlogs) formatted for Microsoft Excel or Google Sheets.
Program Summary Matrix	**CSV Spreadsheet**	Exports consolidated department statistics: Total students, Placed count, FTE/PPO split, Remaining, and SI offer counts.
Raw JSON Intelligence Dump	**JSON Schema**	Exports full raw structured JSON data for API integrations or external Power BI Desktop data model ingestion.

6. Annual Rollover & Template Ingestion Protocol

To update the portal for subsequent graduating cohorts (e.g., Class of 2027), follow this standardized protocol:

Step 1: Download Standard Template: Open the **'Template Manager'** tab and click *'Download Official CSV Template'* to obtain the verified header structure.

Step 2: Prepare Cohort Master Sheet: Populate columns maintaining standard fields: S.No., Enrollment No., Student Name, Department, Gender, CGPA, Backlogs, Summer Internship Status, Placement Status, Company, and CTC (LPA).

Step 3: Upload via Template Manager: Drag-and-drop the updated CSV file directly into the portal. The built-in parser validates column mappings, computes new aggregates, and refreshes all visualizations immediately.

Step 4: Permanent Codebase Update: Run the Python conversion utility: `python generate_json_v2.py` to recompile `spsu_cdc_data.json` for permanent git commits.

7. Hosting Infrastructure & Offline Execution

The portal is architected with dual-availability infrastructure ensuring 100% operational uptime:

Channel	URL / Access Path	Operational Notes
Live Web Portal (Cloud)	https://sarlayash.github.io/SPSU-Career-Development-Cell-CDC-Power-BI-Executive-Intelligence-Portal/	Globally accessible 24/7 over secure HTTPS. Powered by automated GitHub Pages CI/CD from the gh-pages branch.
Offline Standalone File	`standalone_dashboard.html` (in root project directory)	Self-contained 718 KB single-file application. Double-click to run in any browser with ZERO internet connection required.
Source Code Repository	https://github.com/sarlayash/SPSU-Career-Development-Cell-CDC-Power-BI-Executive-...	Official institutional repository containing version history, automated tests, data scripts, and build artifacts.

8. Troubleshooting & Frequently Asked Questions

• **Q: The browser displays an older version of the portal after deployment.**
Fix: Perform a hard cache refresh by pressing **Ctrl + Shift + R** (Windows) or **Cmd + Shift + R** (Mac). The build embeds unique cache-busting hashes.

• **Q: I forgot or need to change the Administrator credentials.**
Fix: Default credentials are **`ADMIN`** / **`SPSU123`**. To update credentials, edit **`src/components/LoginPage.jsx`** and rebuild with **`npm run build`**.

• **Q: How do I export an executive presentation memo without browser URLs or page numbers?**
Fix: In the print preview window, open 'More Settings' and uncheck 'Headers and footers'. The portal's built-in print CSS automatically inserts institutional headers and signature lines.

9. Institutional Authorization & Directorate Sign-Off

This document establishes the official operational standard for the SPSU CDC Executive Intelligence Portal. Any structural modification to database schemas or placement reporting formulas must receive prior approval from the Directorate.

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